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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2025

Sixth Semester

21CSS303T - DATA SCIENCE

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

Note:

- Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- Part - B and Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Marks CO BL

Answer all Questions

- | | | | |
|---|---|---|---|
| 1. Which of the following functions in NumPy is used to create an identity matrix? | 1 | 1 | 1 |
| (A) eye() | | | |
| (B) ones() | | | |
| (C) zeros() | | | |
| (D) diag() | | | |
| 2. In Pandas, which data structure is best suited for handling labeled one-dimensional data? | 1 | 1 | 1 |
| (A) Series | | | |
| (B) DataFrame | | | |
| (C) Panel | | | |
| (D) Matrix | | | |
| 3. If you want to remove missing values from a DataFrame, which method will you use? | 1 | 1 | 3 |
| (A) fillna() | | | |
| (B) dropna() | | | |
| (C) replace() | | | |
| (D) reindex() | | | |
| 4. Which step is NOT part of the Data Science process? | 1 | 1 | 2 |
| (A) Data collection | | | |
| (B) Data visualization | | | |
| (C) Model evaluation | | | |
| (D) Software installation | | | |
| 5. Which Pandas method is used to merge datasets based on a common column? | 1 | 2 | 1 |
| (A) concat() | | | |
| (B) append() | | | |
| (C) merge() | | | |
| (D) join() | | | |
| 6. Why is handling missing data important? | 1 | 2 | 2 |
| (A) It reduces dataset size | | | |
| (B) It prevents model bias and errors | | | |
| (C) It speeds up computation | | | |
| (D) It increases redundancy | | | |
| 7. Which function can be used to detect outliers in a dataset? | 1 | 2 | 2 |
| (A) mean() | | | |
| (B) std() | | | |
| (C) boxplot() | | | |
| (D) reindex() | | | |
| 8. Which transformation is most appropriate before applying machine learning when working with categorical data that are stored as strings? | 1 | 2 | 2 |
| (A) Normalization | | | |
| (B) One-hot encoding | | | |
| (C) Standardization | | | |
| (D) Missing value imputation | | | |
| 9. Which library provides the pairplot() function? | 1 | 3 | 1 |
| (A) matplotlib | | | |
| (B) seaborn | | | |
| (C) pandas | | | |
| (D) plotly | | | |

10. A boxplot is primarily used to: 1 3 1
 (A) Display frequency of categories (B) Show distribution and detect outliers
 (C) Plot linear relationships (D) Visualize correlation
11. Which command in matplotlib saves the plot to a file? 1 3 1
 (A) plt.show() (B) plt.savefig()
 (C) plt.plot() (D) plt.write()
12. A histogram is plotted but the bins are too wide, which hides details of the distribution. Which parameter should be adjusted? 1 3 1
 (A) linewidth (B) alpha
 (C) bins (D) figsize
13. Which of the following is a supervised learning algorithm? 1 4 1
 (A) K-means (B) Linear Regression
 (C) PCA (D) Apriori
14. What is the purpose of splitting the dataset into training and testing sets? 1 4 2
 (A) To reduce computation (B) To evaluate model performance on unseen data
 (C) To increase dataset size (D) To avoid missing values
15. Which algorithm is most appropriate when the dependent variable is categorical? 1 4 2
 (A) Linear Regression (B) Logistic Regression
 (C) K-means (D) PCA
16. A model shows high accuracy on training data but low accuracy on testing data. What does this indicate? 1 4 2
 (A) Underfitting (B) Overfitting
 (C) Normal fitting (D) Good generalization
17. Which algorithm is commonly applied for clustering tasks? 1 5 1
 (A) Linear Regression (B) Logistic Regression
 (C) K-means (D) Decision Tree
18. What is the primary goal of unsupervised learning? 1 5 2
 (A) Predict outcomes (B) Find hidden patterns in data
 (C) Classify data into labeled groups (D) Evaluate supervised models
19. Which method is suitable for dimensionality reduction in datasets? 1 5 1
 (A) KNN (B) PCA
 (C) Random Forest (D) Naive Bayes
20. A clustering algorithm assigns highly dissimilar points to the same cluster. What does this most likely indicate? 1 5 2
 (A) Low variance in data (B) Poor choice of number of clusters
 (C) Correct clustering (D) Overfitting

PART - B (4 × 10 = 40 Marks)

Answer any 4 Questions

Marks CO BL

21. A multinational retail company collects daily transactional data from hundreds of stores across different cities. The data contains product details, quantity sold, price, and timestamp of sales. Management requires the information to be consolidated for identifying best-selling products and analyzing time-based sales trends. Explain how NumPy arrays and Pandas DataFrames can be applied to manage and manipulate this data. Include details of array creation, indexing, slicing, joining, and aggregation. Discuss how this preparation ensures a solid foundation for advanced statistical and predictive analysis. 10 1 3
22. A healthcare organization maintains electronic medical records containing patient demographics, medical history, and diagnostic test results. However, the dataset suffers from missing values in critical fields such as age, inconsistent categorical entries in diagnosis, and extreme outliers in test scores. Develop a structured data wrangling and cleaning process to address these issues. Explain strategies for handling missing data, correcting inconsistencies, and managing outliers. Analyze how these methods ensure the dataset becomes accurate, consistent, and reliable for predictive modeling. 10 2 3
23. A financial analyst is preparing a quarterly report for a global corporation, which includes revenue, expenditure, and profit across multiple regions over several years. Stakeholders demand visualizations that highlight growth trends, regional comparisons, and anomalies. Evaluate the effectiveness of different visualization techniques such as line plots, bar charts, histograms, boxplots, and pair plots using Matplotlib and Seaborn. Provide reasoning on how each visualization communicates insights effectively and supports strategic decision-making. 10 3 3
24. An e-commerce company aims to build a predictive model that classifies whether a customer will purchase a product based on browsing history, demographics, and past purchases. The dataset contains both categorical and numerical variables requiring preprocessing. Apply a supervised learning approach by detailing data preparation steps, model selection (logistic regression, decision trees, or others), and dataset splitting into training and testing sets. Evaluate the effectiveness of the model using accuracy, precision, recall, and F1-score, explaining why these metrics provide a reliable measure of real-world performance. 10 4 3
25. A telecommunications company seeks to segment its customer base to design personalized marketing campaigns. The dataset includes call duration, internet usage, recharge frequency, and complaint history. Analyze how clustering algorithms such as K-means or hierarchical clustering can be applied to identify customer segments. Discuss preprocessing steps, methods for selecting the number of clusters, and evaluation metrics such as silhouette score. Provide an analysis of how effective clustering can improve customer engagement and retention strategies. 10 5 4
26. A social media platform gathers millions of comments daily, containing unstructured text filled with emojis, spelling variations, repeated characters, and irrelevant symbols. A complete text-preparation pipeline is required before conducting sentiment analysis. Create a workflow that includes cleaning, tokenization, stopword removal, normalization, and transformation techniques. Justify why each stage is essential and how the designed pipeline ensures meaningful input for machine learning models. Present a structured framework that could be implemented in practice. 10 2 3

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Question

Marks CO BL

27. A government health department has collected COVID-19 patient records from multiple hospitals over a one-year period. The dataset includes patient ID, age, gender, city, test results, and hospitalization details. However, the dataset is incomplete and inconsistent: several records have missing values in age and gender, some entries show misspelled city names, and test result fields contain multiple representations such as "Positive", "POS", and "+ve". In addition, extreme outliers are found in patient age, such as values above 200. The department requires a cleaned dataset for accurate prediction of hospitalization rates and regional analysis. Data wrangling and cleaning must therefore be performed systematically using Python with Pandas. Identify the different types of data quality issues in the dataset and explain why cleaning is necessary. (5 Marks)

Write the Python syntax to: (10 Marks)

Load the dataset into a DataFrame

Handle missing values appropriately

Standardize inconsistent categorical entries (e.g., city names, test results)

Remove or correct unrealistic outliers in age

28. An insurance company has collected a dataset of customer records containing demographic details, premium paid, number of claims made, and claim settlement history. The management wants to build a predictive model to classify whether a new policyholder is likely to make a claim in the coming year. The dataset includes both categorical variables (gender, policy type) and numerical variables (age, premium, claims). Before building the model, preprocessing such as encoding categorical variables and normalizing numerical values is required. The company is considering supervised learning algorithms such as Logistic Regression and Decision Trees to train the model. The evaluation of model performance should be carried out using metrics like accuracy, precision, and recall ensuring reliability. Evaluate the suitability of Logistic Regression and Decision Trees for this classification problem. (5 Marks)

Write the Python syntax to: (10 Marks)

Import the dataset and preprocess categorical and numerical variables

Split the dataset into training and testing sets

Train a Decision Tree classifier

Compute and display accuracy, precision, and recall scores

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